

Neurological Disorders:-

Psychiatric Disorders with organic / biochemical Etiology:.

Organic psychiatric disorders are psychiatric disorders which are caused by demonstrated pathology of the brain or other organ.

The causes includes:-

- Epilepsy
- Structural disease e.g, brain tumor
- Brain trauma

An organic mental disorder refers to a decrease in mental functioning that is not the result of a psychiatric condition. Sometimes, this term is used interchangeably with the organic brain syndrome(OBS) or chronic organic brain syndrome is now, generally referred to as a neuro- cognitive disorder.

An organic mental disorder can be caused by a brain injury, disease, exposure to toxins or more.

Causes of organic mental disorder:-

Organic mental disorders can have numerous possible causes. Among them are damage or injury to the brain from:

- Cancer
- Chemical or hormonal imbalances
- infection
- Lack of oxygen (anoxic injury)
- Metabolic Syndrome.
- Physical trauma, such as concussion or other head injury
- Vitamin Deficiencies

Symptoms of psychiatric disorder:-

They might have trouble managing relationships and collaborating or communicating with colleagues, friends, or family.

Other symptoms of an organic mental disorders or neuro cognitive disorder include:

- . Agitation
- . Delirium
- . Dementia

Etiology of Psychiatric Disorder:-

There is no single cause for mental illness. A number of factors can contribute to risk for mental illness, such as: Your genes and family history. Your life experiences, such as stress or a history of abuse, especially if they happen in childhood. Biological factors such as chemical imbalances in the brain.

Certain factors may increase your risk of developing a mental illness, including: A history of mental illness in a blood relative, such as a parent or sibling. Stressful life situations, such as financial problems, a loved one's death or a divorce. An ongoing (chronic) medical condition, such as diabetes.

Mood Disorder:-

A mood disorder is a mental health condition that primarily affects your emotional state. It is a disorder in which you experience long periods of happiness, extreme sadness or both. Certain mood disorders involve other persistent emotions, such as anger and irritability.

Mood Disorder includes:-

- . Depression
 - . Bi-polar
 - . Premenstrual dysphoric disorder .
- Disruptive mood dysregulation disorder.

Depression:-

Depression (major clinical depression) is a common mental health condition. Depression symptoms include feeling sad and hopeless. For a person to receive a diagnosis of clinical depression, symptoms must last for at least two weeks.

Persistent depressive disorder:-

This is a chronic form of depression that must last for at least two years.

Depression with psychosis:-

This is a type of severe depression combined with psychotic episodes, such as hallucinations (seeing or hearing things that others don't) or delusions (having fixed but false beliefs). People who experience depression with psychosis have an increased risk of thinking about suicide.

Bipolar Disorder:-

Bipolar disorder is a life mood disorder and mental health condition that causes intense shifts in mood, energy level, thinking patterns and behavior. This are a few types of bipolar disorder, which involve experiencing significant fluctuations in mood referred to as hypomanic / manic and depressive episodes.

Premenstrual dysphoric disorder:-

This type of mood disorder occurs seven to 10 days before menstruation and goes away within a few days of the start of the menstrual period. Researchers believe this condition is brought about by the hormonal changes related to the menstrual cycle. Symptoms may include anger, irritability, anxiety, depression and insomnia.

Disruptive mood dysregulation disorder:-

This disorder affects children and adolescents. It involves frequent anger outbursts and irritability out of proportion to the situation. It occurs before the age of 10.

Anxiety Disorder:-

An anxiety disorder is a type of mental health condition. If you have an anxiety disorder, you may respond to certain things and situations with fear and dread. You may also experience physical signs of anxiety, such as pounding heart and sweating.

Types of anxiety disorders:-

Types of anxiety disorders include:

Generalized anxiety disorder:-

This condition causes fear, worry and a constant feeling. It is characterized by excessive, frequent and unrealistic worries about everyday things, such as job responsibilities, health.

Agoraphobia:-

This condition causes an intense fear of becoming overwhelming or unable to escape or get help. People with agoraphobia often avoid new places and unfamiliar situations, like large, open areas or enclosed spaces, crowds and places outside of their homes.

Panic disorder:-

This condition involves multiple unexpected panic attacks. A main feature of the condition is that the attacks usually happen without warning and aren't due to another mental health or physical condition.

Social anxiety disorder:-

This condition (formerly known as social phobia) happens when you experience intense and ongoing fear of being judged negatively and watched by others.

Symptoms of anxiety Disorder includes:-

- . Feeling panic, fear, or uneasiness
- . Feeling on edge or irritable
- . Un- controllable, obsessive thoughts.

Etiology of Mood Disorder:-

Neurotransmitters, genetics, and psychosocial stressors all seem to play a part in mood disorders.

The same depressed patient may have variable clinical symptoms from one major depressive episode to another. Despite this variability, major depression may have the

same underlying cause. The variable presentations may be due to different patterns in neurotransmitter abnormalities. Deficiencies in serotonin, norepinephrine, dopamine, GABA, and peptide neurotransmitters (somatostatin, thyroid-related hormones, and brain derived neurotrophic factors) have all been hypothesized as to depression. Over activity in other neurotransmitters including substance P, and acetylcholine, and elevated serum cortisol (with lack of diurnal variation) has also been proposed to contribute to depression.

Schizophrenia:-

Schizophrenia is a disabling psychiatric condition impacting around 1% of people worldwide and ranking among the top 10 global disability causes. Schizophrenia is characterized by positive psychotic symptoms such as hallucinations, delusions, disorganized speech, and catatonic behavior.

Negative symptoms such as reduced motivation and expressiveness, cognitive impairments affecting execution function, memory and mental processing speed.

Although current diagnostic and treatment approaches in schizophrenia primarily emphasize psychotic symptoms play a significant role in impairing social and occupational functioning and often show limited response to antipsychotic medications.

Etiology:-

Patients with schizophrenia exhibit both structural and functional brain abnormalities. Lateral ventricle enlargement by about 25% coupled with an approximate 2% reduction in overall brain volume, primarily in the gray matter, is seen in schizophrenia. This decrease in gray matter is noted predominantly in the frontal and temporal lobes and the hippocampus.

Psychosocial stressors in combination with a genetic predisposition have been postulated to alter the size of neurons, neuronal function, repair capabilities, and production of new neurons.

Elevated cortisol in some depressed patients may reduce hippocampal volume, especially if their depression has not been treated in some time. Brain imagery has also noted some altered structures, which suggest some changes in neurocircuitry.

Psychosocial theories suggest that experiences of “loss” in certain vulnerable individuals may cause depression, either through trauma, parental loss, loss of love from others, or loss of self-esteem.

Mental Retardation:-

Mental retardation (or intellectual disability) means having below average intelligence and skills necessary for daily life. It exists in children and adults whose brains do not develop properly or function within the normal range.

Intellectual reasoning or I.Q. (Intelligence Quotient), usually measured by a standardized test): This is the ability to think, reason, learn, make decisions, and solve problems.

The average person's IQ is 100. Anyone with an IQ below (70) or (75) is considered intellectually disabled.

Adaptive behaviors:- This is the area that includes necessary life skills such as socialization, communication, and taking care of oneself.

Genetics Disorders:- Many genetic disorders can cause mental retardation. Examples include Down syndrome, Fragile X syndrome, Prader-Willi syndrome, and Williams syndrome. These disorders involve specific genetic mutations or chromosomal abnormalities that affect brain development and functioning, leading to intellectual disabilities.

Metabolic Disorders:- Certain metabolic disorders can result in mental retardation. For instance, phenylketonuria (PKU) is a genetic disorder in which the body cannot properly process the amino acid phenylalanine. The build-up of phenylalanine can cause brain damage and intellectual disability if left untreated.

Prenatal and Perinatal Factors:- Various factors occurring during pregnancy or childbirth can contribute to mental retardation. These include maternal infections, exposure to toxins or drugs, severe malnutrition, maternal substance abuse,

complications during pregnancy or delivery, and lack of oxygen to the baby's brain (birth asphyxia).

Brain Injury or Damage:- Traumatic brain injury, infections of the brain (such as meningitis or encephalitis), brain tumors, stroke, or other conditions that cause damage to the brain can result in mental retardation.

Biochemical Imbalances:- Some biochemical imbalances can lead to intellectual disabilities. For example, untreated hypothyroidism (under- active thyroid) in infancy or childhood can cause mental retardation. Imbalances in neurotransmitters or disruptions in specific biochemical pathways can also contribute to cognitive impairments.

It is important to note that mental retardation can result from a combination of a genetic environmental and biochemical factors and it is often challenging to identify a single cause in many cases. Diagnosis and management of mental retardation involve comprehensive assessment including genetic testing, medical evaluation, and assessments of cognitive and adaptive functioning. Treatment approaches focus on supportive intervention, educational and behavioral therapies, and addressing any underlying medical conditions or biochemical imbalances to optimize functioning and quality of life for individuals with mental retardation.